

Figure 4 — Within-channel sentence-vs-token cost-per-correct on non-floor non-anomalous cells.
Sentence-grouping is cheaper per correct within every plotted cell: $\sim 1.25\text{-}2.72\times$ on niah primary, $\sim 1.76\text{-}4.03\times$ on qa_2 cross-task.
Excluded: Gemma 4 vorn niah (token floor 1/50, ratio unstable); Gemma 4 token-H2O niah (cost anomaly \$2.18, suspected cold-start);
Gemma 4 vorn qa_2 (token floor 10/200, parallel to niah floor exclusion).

